LETTERS: NEW OBSERVATIONS

Potential *PINK1* Founder Effect in Polynesia Causing Early-Onset Parkinson's Disease

We present a pilot study investigating PINK1 variant early-onset Parkinson's disease (EOPD; age of onset before 51 years[1]) in Pacific people in New Zealand. In 2009, one Samoan and one (unrelated) Tongan man from Auckland, New Zealand with EOPD were found to be homozygous for the same rare pathogenic variant: NM_032409.3(*PINK1*):c.1040 T > C p.(Leu347Pro) rs28940285 (*PINK1*:c.1040 T > C). The variant's disease-causing status has been established from being found in a homozygous state in small groups of EOPD patients from the Philippines, Guam, and Malaysia.[2-4]

We recruited Māori or Pacific participants with EOPD in New Zealand; Pacific patients with late-onset PD were recruited as controls (Health and Disability Ethics approval HDEC 13/NTA/230 AM01). Some 373 healthy Māori or Pacific controls were also genotyped for the *PINK1*:c.1040 T > C variant (Health and Disability Ethics approval MEC/05/10/130).

Five Pacific EOPD patients were recruited (two Samoan and three Tongan, no known relatedness; three male and two female; average age of onset 33 ± 10 years). All five were found to be homozygous for the *PINK1*:c.1040 T > C variant. Four reported a family history of PD (Table S1).

The three late-onset patients (one Tongan, one Samoan, one Māori; two female and one male; average age of onset 57 ± 4 years) were negative for the *PINK1*:c.1040 T > C variant.

All five EOPD patients had typical dopa-responsive PD. Three reported lower limb dystonia at presentation or early in their disease course and four reported neuropsychiatric symptoms (eg, low mood or anxiety) at presentation or prior. Montreal Cognitive Assessment scores suggested cognitive impairment in three individuals.

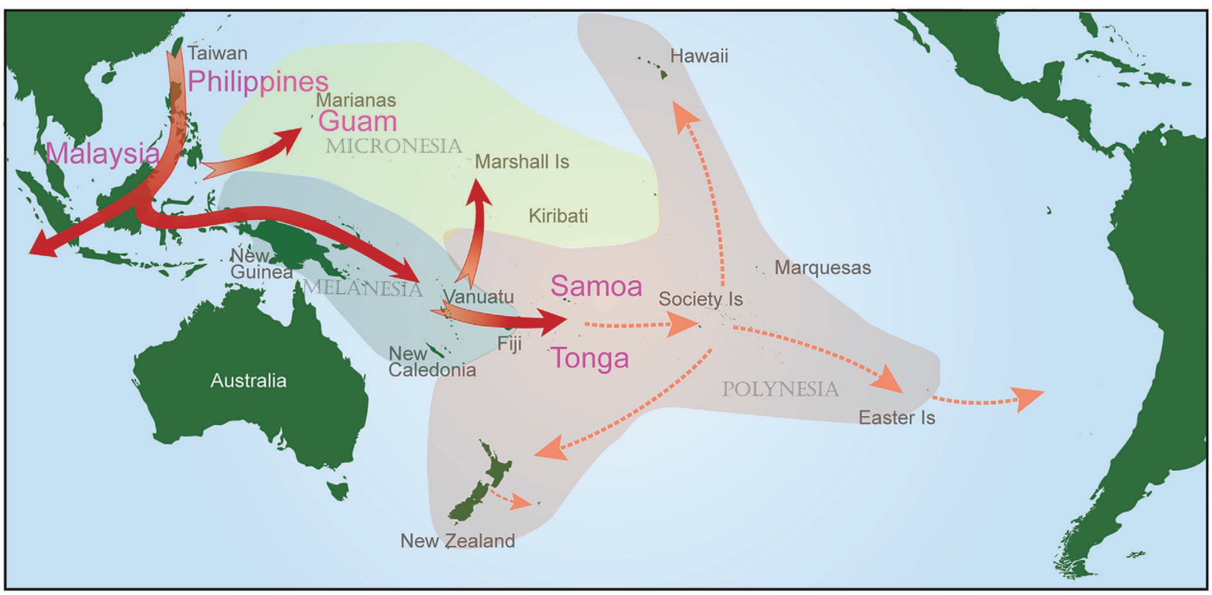

FIG. 1. Postulated migration routes from maritime South East Asia into the Pacific. Migrations 3000-3500. years-before-present into West Polynesia (red arrows); migrations 730-1200 years-before-present into East Polynesia (orange-dotted arrows). Countries where populations carry *PINK1*: c.1040T>C (pink). Map adapted with permission from.[5]

Key Words: early onset Parkinson's disease; recessive genetic conditions; *PINK1*; dystonia; Polynesia

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Of the 373 Māori and Pacific healthy controls, 273 were genotyped successfully (73%). Eight were heterozygous for the *PINK1*:c.1040 T > C variant (no homozygotes), giving an allele frequency of 1.5%. Seven were amongst the 126 West Polynesian participants (allele frequency of 2.8%) and one (who identified as Māori) was amongst the 137 East Polynesian participants (allele frequency 0.4%).

All seven West Polynesian patients with EOPD (two index patients and five study participants) were homozygous for the same *PINK1* variant, *PINK1*:c.1040 T > C, which has only previously been reported in the Western Pacific. This is in marked contradistinction to the global frequency of PINK1 mutations, which are rare. Given the history of human migrations in the Pacific[5] it is likely that the variant was brought during migration, and that there has been a founder effect in the peoples of West Polynesia (Samoa, Tonga, Tokelau, and Niue). The variant may have continued into the Eastern Polynesian migrations including Hawaii, Cook Islands, and New Zealand Māori (Fig. 1). An allele frequency of 2.8% in a sample of West Polynesians may have significant health implications as this could imply an EOPD prevalence rate due to *PINK1*:c.1040 T > C homozygosity of ~1 in 5000 people in these populations (Appendix S1).

Our cohort had clinical phenotypes which were consistent with the limited description of PINK1 EOPD in the literature.[6]

We intend to extend our patient and population studies to further characterize and quantify our findings in the wider Pacific population in New Zealand.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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References

1. Alcalay RN, Caccappolo E, Mejia-Santana H, et al. Frequency of known mutations in early-onset Parkinson disease: implication for genetic counseling: the consortium on risk for early onset Parkinson disease study. Arch Neurol 2010;67(9):1116–1122.
2. Rogaeva E, Johnson J, Lang AE, et al. Analysis of the PINK1 gene in a large cohort of cases with Parkinson disease. Arch Neurol 2004;61 (12):1898–1904.
3. Steele JC, Guella I, Szu-Tu C, et al. Defining neurodegeneration on Guam by targeted genomic sequencing. Ann Neurol 2015;77(3):458–468.
4. Tan AH, Lohmann K, Tay YW, et al. PINK1 p.Leu347Pro mutations in Malays: prevalence and illustrative cases. Parkinsonism Relat Disord 2020;79:34–39.
5. Matisoo-Smith E. Ancient DNA and the human settlement of the Pacific: a review. J Hum Evol 2015;79:93–104.
6. Kilarski LL, Pearson JP, Newsway V, et al. Systematic review and UK-based study of PARK2 (parkin), PINK1, PARK7 (DJ-1) and LRRK2 in early-onset Parkinson's disease. Mov Disord 2012;27(12):1522–1529.

Supporting Data

Additional Supporting Information may be found in the online version of this article at the publisher's web-site.

Eye-of-Tiger Sign in Globus Pallidus: A Novel Radiological Feature of Spinocerebellar Ataxia Type 28

Spinocerebellar ataxia type 28 (SCA28) is a rare autosomal-dominant, neurodegenerative disorder characterized by progressive cerebellar ataxia and variable neurological signs.[1] SCA28 is caused by heterozygous mutations in the ATPase family gene 3-like 2 (*AFG3L2*) gene.[2] Notably, homozygous mutations in *AFG3L2* cause autosomal-recessive spastic ataxia 5.[3] *AFG3L2* encodes a mitochondrial ATP-dependent metalloprotease, which is highly expressed in

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